

Pathophysiology and antioxidant status of patients with fibromyalgia.

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Fibromyalgia syndrome (FMS) is characterized by systemic pain of unknown etiology and is often accompanied by various psychological symptoms. Research on different parameters in fibromyalgia (FM) indicates that multifactors are involved in its pathophysiology; such as genetic factors, substance P, serotonin, hypothalamic pituitary adrenal axis (HPA), muscles metabolic dysfunction, reactive oxygen species (ROS) and reactive nitrogen species (RNS). Oxidative stress has also been implicated in the pathophysiology of FM; therefore, supplementation with antioxidants may be important in modulation of the effects of ROS in patients with FM.

[Viral infection and fibromyalgia].

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Fibromyalgia (FM) is a very frequent syndrome of unknown cause, characterized by generalized pain, fatigue and a number of tender points to palpation. Among the several etiopathogenic hypotheses discussed, the association of FM with some viral infections has been the object of multiple studies due to its relation and similarity with the chronic fatigue syndrome, acknowledges as being related, although not exclusively, with the chronic infection by the Epstein-Barr virus. Many individual descriptions of association between infection with the human parvovirus B19 and FM led us to carry out this study, comparing the serology for that virus in 52 patients with FM and 39 healthy controls. The titers of specific IgG anti-parvovirus B19 antibodies, indicating previous infection with that virus, were determined in all 91 individuals through ELISA method and at the same laboratory. Results revealed, though not significantly, a greater prevalence of positive titers, of which the mean was also higher, in patients than in controls. When comparing the women from both groups, this tendency was even less perceptible. These data imply that there is no etiologic association between infection with the human parvovirus B19 and FM.

Influence of Oxidative Stress-Related Genes on Susceptibility to Fibromyalgia.

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Fibromyalgia (FM) is a complex syndrome to diagnose and treat because of its unknown etiology. However, previous studies reported that patients with FM experience oxidative stress. In this study, we investigated single-nucleotide polymorphisms (SNPs) in genes encoding enzymes involved in oxidative stress (superoxide dismutase 1 [SOD1], catalase, and NADPH oxidase [CYBA]) in patients with FM and in healthy subjects, as well as the possible relation with demographic and clinical manifestations of FM. A total of 141 patients with FM and 73 healthy subjects participated in this case-control study. For DNA extraction, buccal swabs were collected from patients with FM, and a peripheral blood sample was extracted from controls. We analyzed SNPs in genes related to oxidative stress (rs10432782 in SOD1, rs1001179 in catalase, and rs4673 in CYBA) using TaqMan probes. In patients with FM, severity of FM, fatigue, and pain were assessed by Fibromyalgia Impact Questionnaire, Multidimensional Fatigue Inventory, and Visual Analogue Scale (VAS), respectively. Physical (PCS-12) and mental (MCS-12) health statuses were evaluated by the 12-Item Short-Form Health Survey. The selected SNPs did not show

significant differences between patients with FM and controls. The rs10432782 (SOD1) was associated with Fibromyalgia Impact Questionnaire scores in patients with FM, whereas the rs4673 (CYBA) was associated with the Multidimensional Fatigue Inventory score, MCS-12 score, and duration of the disease. We have identified significant correlations between SOD1 and CYBA variants with clinical manifestations of FM. These results provide new insights into the pathogenesis of FM that could be useful for guiding future studies along the way to find the cause(s) of this syndrome.

Fibromyalgia syndrome: an overview of potential drug targets.

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Fibromyalgia syndrome (FMS) is a chronic disease of widespread and debilitating pain. The cause of FMS is unknown and its risk factors are poorly understood. It occurs frequently in the general population where it is often co-morbid with other rheumatoid and pain disorders, and psychiatric disorders such as anxiety and depression, making diagnosis particularly difficult. Several types of drugs are used to treat FMS, but none are specifically approved for this indication. FMS appears to be strongly associated with depression or at least with some symptoms of depression, and antidepressants appear to be effective in the treatment of this disorder. The advent of new classes of antidepressants with fewer side effects than older drugs has suggested new avenues of therapy for patients diagnosed with FMS.

Fibromyalgia syndrome: etiology, pathogenesis, diagnosis, and treatment.

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Fibromyalgia syndrome is mainly characterized by pain, fatigue, and sleep disruption. The etiology of fibromyalgia is still unclear: if central sensitization is considered to be the main mechanism involved, then many other factors, genetic, immunological, and hormonal, may play an important role. The diagnosis is typically clinical (there are no laboratory abnormalities) and the physician must concentrate on pain and on its features. Additional symptoms (e.g., Raynaud's phenomenon, irritable bowel disease, and heat and cold intolerance) can be associated with this condition. A careful differential diagnosis is mandatory: fibromyalgia is not a diagnosis of exclusion. Since 1990, diagnosis has been principally based on the two major diagnostic criteria defined by the ACR. Recently, new criteria have been proposed. The main goals of the treatment are to alleviate pain, increase restorative sleep, and improve physical function. A multidisciplinary approach is optimal. While most nonsteroidal anti-inflammatory drugs and opioids have limited benefit, an important role is played by antidepressants and neuromodulating antiepileptics: currently duloxetine (NNT for a 30% pain reduction 7.2), milnacipran (NNT 19), and pregabalin (NNT 8.6) are the only drugs approved by the US Food and Drug Administration for the treatment of fibromyalgia. In addition, nonpharmacological treatments should be associated with drug therapy.

Update on fibromyalgia therapy.

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Primary fibromyalgia, a poorly-understood chronic pain syndrome, is characterized by widespread musculoskeletal pain, nonrestorative sleep, fatigue, psychological distress, and specific regions of localized tenderness, all in the absence of otherwise apparent organic disease. While the etiology of fibromyalgia is unclear, accumulating data suggest that disordered central pain processing likely plays a role in the pathogenesis of symptoms. Although various pharmacological treatments have been studied and espoused for treating fibromyalgia, no single drug or group of drugs has proved to be particularly useful in treating fibromyalgia patients as a whole, and only one drug to date has earned U.S. Food and Drug Administration approval for treating the syndrome in the United States. This review critically and systematically evaluates clinical investigations of medicinal and nonmedicinal treatments for fibromyalgia dating from 1970 to 2007.

Fibromyalgia syndrome is associated with hypocortisolism.

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Fibromyalgia syndrome (FMS) is a disease of unknown pathogenesis characterized by chronic musculoskeletal pain. FMS has been also associated with altered endocrinological responses, but findings are inconsistent. The aim of the present study was to investigate free salivary cortisol levels in FMS patients compared with healthy controls with a particular focus on the cortisol awakening response (CAR). The saliva samples were collected in a controlled hospital-hotel setting, in which the participants' compliance was high and a number of potential confounders were analyzed. Twenty-nine chronic female FMS patients and 29 age-matched healthy female controls were recruited. Salivary cortisol samples were investigated eight times: in the afternoon when participants arrived at the hospital, after stress provocation (to be reported separately), in the evening, before they went to sleep, upon awakening, 30 and 60 min later, and during the afternoon of the second day. Questionnaires measuring pain levels, sleeping problems, perceived stress, and personality were administered to the participants. Other psychophysiological measurements were used to assess sleep quality and heart rate. Patients with FMS had significantly lower cortisol levels during the day, most pronounced in the morning (CAR). The potential confounders analyzed did not influence the results. As expected, FMS patients reported more pain, stress, sleeping problems, anxiety, and depression. The results lend support to the hypothesis of a dysfunction in the hypothalamus-pituitary-adrenal axis in FMS patients, with generally lower cortisol values, most pronounced upon awakening (CAR).

Exploring the role of biologics in depression.

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Depression is a chronic and prevalent neuropsychiatric disorder; clinical symptoms include excessive sad mood, anhedonia, increased anxiety, disturbed sleep, and cognitive deficits. The exact etiopathogenesis of depression is not well understood. Studies have suggested that tumor necrosis factor- α (TNF- α) and interleukins (ILs) perform vital roles in the pathogenesis and treatment of depression. Increasing evidence suggests the upregulation of TNF- α and ILs expression in patients with depression. Therefore, biologics like TNF inhibitors (etanercept, infliximab, adalimumab) and IL inhibitors (ustekinumab) have become key compounds in the treatment of depression. Interestingly, treatment with an antidepressant has been found to decrease the TNF- α level and improve depression-like behaviors in several preclinical and clinical studies. In the current article, we have reviewed the recent findings linking TNF- α and the

pathogenesis of depression proving TNF- α inhibitors as potential new therapeutic agents. Animal models and clinical studies further support that TNF- α inhibitors are effective in ameliorating depression-like behaviors. Moreover, studies showed that peripheral injection of TNF- α exhibits depressive symptoms. These symptoms have been improved by treatment with TNF- α inhibitors. Hence suggesting TNF- α inhibitors as potential new antidepressants for the management of depressive disorder.

What have we learnt from the inhibition of IL-6 in RA and what are the clinical opportunities for patient outcomes?

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Rheumatoid arthritis (RA) is an autoimmune disease characterised by persistent inflammation of the synovial joints as well as other tissues and organs. Left untreated, it can lead to joint damage, disability and even increased mortality. The disease is driven by inflammatory cytokines that contribute to the chronic inflammation seen in RA. Interleukin-6 (IL-6) is a key pathological cytokine and a target for treatments aiming to alleviate local and systemic inflammation. Despite advances in understanding RA and the introduction of new treatments, achieving sustained remission remains challenging. This review explores the role of IL-6 in RA pathogenesis, its potential as a treatment target and the significance of personalised medicine in RA management. IL-6 has a dual signalling mechanism, classical and trans-signalling, which influences various intracellular pathways. While several targeted therapies have emerged, no single mechanism-based therapy is universally effective due to the diversity and complexity of the disease. Different approaches to targeting IL-6 have been tested, including biologic blockade of receptors or ligands, and inhibition of IL-6 signalling. IL-6 receptor inhibitors have been validated as RA therapeutics, either alone or in combination with other treatments. Tocilizumab, the first approved IL-6 inhibitor, blocks both soluble and membrane-bound IL-6 receptors, reducing the inflammatory cascade. Clinical trials confirm the efficacy and safety of tocilizumab and its role as a treatment option for patients unresponsive to conventional therapies. The benefits of IL-6 inhibition extend beyond reduced joint inflammation to the amelioration of comorbidities like anaemia, cardiovascular disease, depression and osteoporosis. Tailoring treatment to patients' profiles and comorbidities is essential for optimal outcomes. A 'treat-to-profile' approach, focusing on a holistic view of the patient, could improve personalised medicine strategies. Biosimilars - lower-cost alternatives to biologics - further enhance the accessibility and cost-effectiveness of treatment. IL-6 inhibitors present a valuable treatment option for RA management, particularly for patients with specific comorbidities.

BioPAXViz: a cytoscape application for the visual exploration of metabolic pathway evolution.

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BioPAXViz is a Cytoscape (version 3) application, providing a comprehensive framework for metabolic pathway visualization. Beyond the basic parsing, viewing and browsing roles, the main novel function that BioPAXViz provides is a visual comparative analysis of metabolic pathway topologies across pre-computed pathway phylogenomic profiles given a species phylogeny. Furthermore, BioPAXViz supports the display of hierarchical trees that allow efficient navigation through sets of variants of a single reference pathway. Thus, BioPAXViz can significantly facilitate, and contribute to, the study of

metabolic pathway evolution and engineering. BioPAXViz has been developed as a Cytoscape app and is available at: <https://github.com/CGU-CERTH/BioPAX.Viz>. The software is distributed under the MIT License and is accompanied by example files and data. Additional documentation is available at the aforementioned GitHub repository. ouzounis@certh.gr.
